

Product Requirements

Graph Generation with Sonify

Team 6

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Revision History

Date	Version	Description	Author
9-17-2024	1.0	Software requirements.	Team 6
11-23-2024	1.1	Updates to reflect sponsor feedback	John Whitgrove
12-08-2024	1.2	Updates on user stories.	Sain Madrigal

Problem Statement

This website is meant to provide an easy, efficient way to load data and produce responsive graphs including sonification. Accessibility will be a top priority, as it will be made with people with disabilities in mind. There will be functionalities such as editing, different graphing options, sound options, and personalization within the website itself. Furthermore, it'll include exportation of the graphs after creation, as well as different modes to accommodate the visually and sonically impaired. The main goal of this project is to create a fast, efficient way of graph creation that can be used by everyone, including impaired individuals from any background.

We are initially going to implement line graphs, box plots, and bar graphs into this web application, with room for expansion.

Stakeholders

End Users – Visually Impaired and Sighted College Students

Individuals who will use the website to enter or load data and receive graph outputs with sound. This includes users who are blind or visually impaired and rely on screen readers, as well as sighted users who may benefit from the auditory representation of data.

Data Analysts

Professionals who use the graphs generated by the website to interpret data. They may provide feedback on the usability and functionality of the graphs and the effectiveness of the sonification features.

Software Engineering Team

Individuals who are involved in designing, coding, and maintaining the website. They are responsible for implementing the features of the site, including the graph generation and sound output functionalities. Some responsibilities include:

- Develop and integrate graph generation and sonification features.
- Ensure the website is responsive and functions correctly.
- Address and fix any technical issues that arise during development and after deployment.

Project Sponsor

Individual providing sponsorship for the project. They provide specific requirements and expectations related to the project's scope and outcomes. Below are some responsibilities:

- Monitor the project's progress and ensure it aligns with their criteria.
- Collaborate with Software Engineering Team and provide additional support as needed.
- Review and approve project deliverables and outcomes.

System requirements

The project shall adhere to the following requirements:

- The project shall use JavaScript as the programming language
- The project shall use the Tone.js library to achieve sonification features.
- The project shall use the chart.js library to achieve data representation features.
- The project shall be compatible with any modern web browser.
- ~~- The project shall use a database to store and process user data.~~

Users profile

- Visually Impaired Users:
 - Being able to distinguish sounds. The user should be able to distinguish between different sounds created based on the graph data to try and visualize it.
 - Basic graph knowledge. The user should know how different graphs work and how each differs from others.
 - Basic knowledge of data tools.
- Researchers:
 - Intermediate reading level.
 - Basic graph knowledge. The user should know how different graphs work to try and visualize them. The user should also know how to create them.
 - Be an expert with data tools. The user should be able to handle data correctly and know how to put that data in a graph.
- College Students:
 - Intermediate reading level.
 - Basic graph knowledge. The user should know how different graphs work to try and visualize them. The user should also know how to create them.
 - Moderate knowledge of data tools. The user should at least understand how to handle data correctly and know how to put that data in a graph.

Functional requirements (user stories)

No.	User Story Name	Description	Priority
R1	Upload Data	Upload a CSV file to the system. This file will be used to generate a graph stored locally on the browser.	1
R2	Manual data Input	Optional selection for manual entry of graph data. Will be limited in scale comparative to CSV entry.	2
R3	Generate Graph	Create representative graph from user-input data and custom name. Graph type is selectable to determine how the data is represented.	1
R4	Sonify Generated Graph	Create an auditory representation of the graph and its data. This will occur in tandem with the generation of the graph. The sonification will include controls and setting to tune to user needs and preferences.	1
R5	Export generated Graph	Export generated graph as a PNG image via a selectable menu item or through directly saving the image.	2
R6	Graph Type Customization	Enable users to customize the type of graph (bar, line, scatterplot) and its appearance (colors, labels) based on their preferences.	2
R7	Accessible Documentation	Provide a help section or tutorial that assists users in understanding how to use the platform and its sonification features.	2
R8	Export Sonification Data	Export the generated graph with sonification as a video file. This will be done via an option on the page displaying the graph.	3

Non-Functional Requirements

1. Reliability

- 1.1. The system shall be available 99.9% of the time during normal operations.
- 1.2. The system shall have reliable error handling for invalid or improperly formatted data input.

2. Usability

2.1. User Interface

- 2.1.1. The system shall offer a consistent and uniform look across all pages.
- 2.1.2. The system shall be intuitive, allowing users to easily navigate between uploading data, generating graphs, and accessing sonification options.
- 2.1.3. The system shall provide clear and actionable error messages when issues arise (file upload errors, incorrect data input).

2.2. Accessibility

- 2.2.1. The system shall comply with WCAG standards to ensure full accessibility for users with disabilities.
- 2.2.2. The system shall support screen readers by providing accessible labels, roles, and descriptions for all interactive elements.

3. Cross-Platform Compatibility

- 3.1. The system shall be compatible with all major browsers, including Chrome, Firefox, Safari, and Edge.
- 3.2. The system shall support different screen resolutions, providing a responsive design that adjusts appropriately to different devices.

4. Accuracy

- 4.1. The system shall ensure the accuracy of sonified data by using clear, distinct sounds that correctly map to data points on the graph.
- 4.2. The system shall validate all input data to ensure that only properly formatted CSV files or manual data entries are accepted.
- 4.3. The system shall generate graphs that accurately represent the inputted data, providing clear visual and auditory feedback.

5. Performance

- 5.1. The system's performance shall be largely dependent on the client's internet connection and hardware capabilities.

6. Security

- 6.1. The system shall implement measures to prevent common vulnerabilities such as Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF) attacks.

7. Maintainability

- 7.1. The system shall be modular to allow future developers to easily update the sonification or graph generation features without impacting other components.
- 7.2. The system's code shall be well-documented, ensuring that future developers can maintain and extend the system without extensive onboarding.

8. Scalability

- 8.1. The system shall be scalable to handle an increased number of users or larger file sizes without a noticeable impact on performance.
- 8.2. The system shall be designed to support additional features, such as more sonification styles or data storage between sessions, with minimal refactoring.

Use Case Diagram

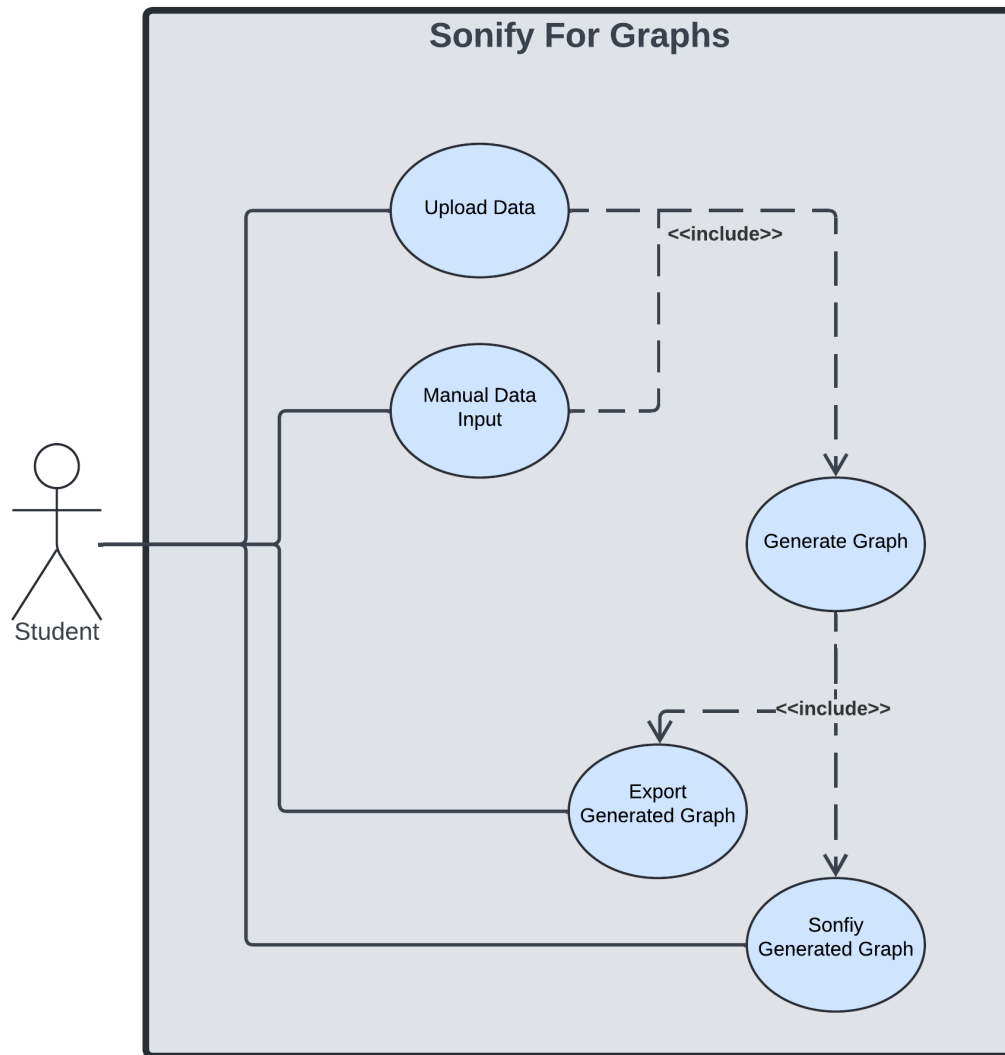


Figure 1: Use Case Diagram

Use Case Description

Use Case Number	UC-01
Use Case Name	Upload Data
Overview	The user shall upload a CSV file to the system, which will be used to generate the graph.
Actor(s)	College Student (sighted and visually impaired)
Pre-Condition(s)	• The website is running and accessible. • The user has accessed the website and is on the data upload page. • The user has a valid CSV file.
Scenario Flow	Main (Success) Flow: 1. The user selects the option to upload a file. 2. The system opens a file browser window for the user to select a CSV file. 3. The user selects a CSV file and clicks the “Open” button. 4. The system verifies the file format and uploads the data. a. If the file is invalid (incorrect format or too large), the system will display an error message and return to Step 2. b. If the file is valid, the system will parse the data and move to the graph generation stage.
Alternate Flow	• If the user selects the wrong file format, the system will display an error message and prompt the user to upload a CSV file. • The user can cancel the upload by clicking the “Cancel” button, and the system will return to the main page.
Post Condition	• The system will store the data locally on the user’s browser for graph generation.

Use Case Number	UC-02
Use Case Name	Manual Data Input
Overview	The user shall input data manually through a form.
Actor(s)	College Student (sighted and visually impaired)
Pre-Condition(s)	• The website is running and accessible. • The user has accessed the manual data input page.
Scenario Flow	Main (Success) Flow: 1. The user selects the option for manual data input. 2. The system displays a form for the user to data (e.g., X and Y values). 3. The user fills out the form and clicks the “Submit” button. 4. The system verifies the data is in the correct format. a. If the data is invalid (non-numeric values), the system will display an error message and return to Step 2. b. If the data is valid, the system stores the data locally for graph generation.
Alternate Flow	• The user can cancel the manual data input process at any time, and the system will return to the main page.
Post Condition	• The system will store the manually entered data locally on the user's browser for graph generation.

Use Case Number	UC-03
Use Case Name	Generate Graph
Overview	The system shall generate a visual graph based on the uploaded or manually entered data. The user shall be able to enter a custom name for the graph.
Actor(s)	College Student (sighted or visually impaired)

Pre-Condition(s)	- The user has uploaded a valid CSV file or entered data manually. - The system has successfully parsed the data. - The user has inputted a custom name for the graph.
Scenario Flow	Main (Success) Flow: - The user selects the option to generate a graph. - The system prompts the user to choose a graph type (e.g., line graph, bar chart). - The system prompts the user to input a custom name for the graph (e.g., Age Graph). - The user selects a graph type, enter custom name, and clicks "Generate." - The system generates the graph based on the input data, name, and displays it on the screen.
Alternate Flow	If the user selects an invalid graph type for the data, the system will display an error message and prompt the user to select a valid type.
Post Condition	The system displays the generated graph on the screen.

Use Case Number	UC-04
Use Case Name	Sonify Generated Graph
Overview	The system shall create an auditory representation of the generated graph.
Actor(s)	College Student (sighted and visually impaired)
Pre-Condition(s)	The user has successfully generated a graph.
Scenario Flow	Main (Success) Flow: - The user selects the option to sonify the graph. - The system generates an auditory representation of the graph.

	- The system plays the sound to the user as the graph is displayed. - The user selects sonify settings (e.g., Play, Stop, Speed).
Alternate Flow	If the graph data is too large or complex, the system will provide an error message indicating that sonification may not be optimal.
Post Condition	The system provides auditory feedback representing data points on the graph.

Use Case Number	UC-05
Use Case Name	Export Generated Graph
Overview	The system shall export the generated graph as a PNG image.
Actor(s)	College Student (sighted and visually impaired)
Pre-Condition(s)	The user has generated a graph.
Scenario Flow	Main (Success) Flow: - The user selects the option to export the graph. - The system provides a downloadable PNG of the graph. - The user downloads and saves the image file to their device.
Alternate Flow	If there is an issue with file export, the system will display an error message.
Post Condition	The user successfully exports the graph as a PNG image.

Contributors

Dillon Edington – Systems Requirements

Sain Madrigal – User Profiles

Isaac Rea – Stakeholders, Non-Functional Requirements, Use Case Diagram, Use Case Description

John Whitgrove – Functional Requirements

Katie Williams -- System Requirements

Dillon Wright – Problem Statement

Sponsor Requirements

I have read and approved the material in this document. If there is no external sponsor, the TA or instructor will sign it for accuracy/scope.

Print Name

Signature

Date

